Blood Work Analysis — Roman

Draw of June 8, 2026 | Report generated June 9, 2026

Sources: yesterday's draw, 10-yr consolidated history, active 31-compound protocol & 30-day adherence, Function Health panel, and parallel literature review (PubMed + ClinicalTrials.gov)

> ■ **Not medical advice.** This is a data + literature synthesis to bring to your physician. Several panels (iron, testosterone) were still "in progress" at report time, so conclusions are preliminary.

> **TL;DR.** A reassuring panel with a genuinely excellent trend story. The four flags are either **longstanding / constitutional** (macrocytosis) or **clinically trivial** (trace leukocyte esterase). Nothing acute. The headline is in the *trends*, not the flags — and much of it is plausibly attributable to your protocol, in the direction you'd want.

1. What flagged yesterday (Jun 8)

Marker	Value	Range	Flag	Verdict
RBC	4.02	4.20–5.80	LOW	Longstanding mild — see §3A
MCV	102.5	81.4–101.7	HIGH	Constitutional, stable since 2016
MCH	33.6	27.0–33.0	HIGH	Rides along with MCV
Leukocyte esterase (urine)	TRACE	Negative	flag	Clinically minor — see §3D

Everything else was clean: WBC 5.4, platelets 250, full differential normal, Hgb 13.5, Hct 41.2, RDW 14.4; urine protein / blood / glucose / nitrite all negative; **ferritin 62** (normal). Iron/TIBC/%sat and the testosterone panel were still **in progress** — get those before drawing firm conclusions.

2. The trend story (the headline)

Your cardiometabolic profile has transformed over ~6 months:

Marker	Dec 2025	Latest 2026	Δ
Total cholesterol	220	**141**	–79
LDL	135	**77**	–58
ApoB	97	**69**	–28
Triglycerides	128	**37**	–91
Non-HDL	148	**89**	–59
hs-CRP	0.6	**0.2**	very low inflammation
Insulin (uIU/mL)	2.5	2.8	excellent sensitivity
HbA1c (%)	5.4	5.3	optimal
Testosterone total (ng/dL)	261 (Sep'25)	616 (Apr'26)	optimized off low baseline
Free T (pg/mL)	33.1	48.6	now in range

Liver enzymes stable (AST mildly high at 44 — almost certainly **muscle/training source** given heavy IGF-1/resistance work; ALT normal at 29). Kidney function stable (eGFR ~79–95). Two minor trade-offs: HDL drifted down (76 → 52) and Lp(a) read lower (88 → 58) — both explained below; neither is a concern.

3. Can it be attributed to peptides? — the core question

Each finding was tested against the literature. Bottom line: the favorable trends are well-explained by your stack, and the "concerning" CBC findings are explained by it too — in a protective way.

A) The low-RBC-despite-TRT paradox → most likely GH-axis hemodilution

On ~140 mg/wk testosterone you'd *expect* high hematocrit. Instead RBC is mildly low and Hct normal. Two opposing forces:

- **GH secretagogues (Tesamorelin + Ipamorelin) and IGF-1 LR3 expand plasma/extracellular fluid volume** (sodium/water retention), diluting concentration-based indices — a documented GH-class effect.
- **Daily SubQ micro-dosed testosterone is the least erythrocytosis-prone route** — steady, no supraphysiologic peaks. Peak-driven IM ~triples erythrocytosis risk.

Møller, JCEM 1991 (PMID 1826008); tesamorelin RCTs Falutz PMID 20101189, Stanley JAMA 2014; Madsen, JCEM 2021 (PMID 33599731); Wilson, AJHP 2018 (PMID 29367424).

Net: the GH/IGF axis is masking the erythrocytosis TRT would otherwise cause — protective (lower thrombosis risk). With normal-for-you MCV, replete iron, and a stable 9-month trend, this is **not a true anemia**. **Flip side:** if GH agents are tapered, watch for Hct rising into erythrocytotic range. **Confirm with a reticulocyte count** (± LDH/haptoglobin).

B) Macrocytosis (MCV/MCH) → constitutional, NOT peptide-driven

MCV has been ~98–102 on **every draw since 2016** — it predates the entire stack. B12 normal (485–642), no anemia, only mildly elevated (vs the 109–113 typical of megaloblastic/MDS). Classic **constitutional macrocytosis**. No human evidence links your NAD-pathway compounds (NAD+, 5-Amino-1MQ, 6-methoxynicotinamide, ATX-304, SS-31, MOTS-c) to raised MCV, and the timing (macrocytosis first) argues against causation. **Confirm benign** with reticulocyte, RDW (already 14.4, reassuring), peripheral smear, MMA + homocysteine, RBC folate, TSH, and an alcohol-intake check. If normal → no marrow biopsy, nothing to do. *(Colon-Otero, Med Clin North Am 1992, PMID 1578958.)*

C) The lipid transformation → yes, attributable to Retatrutide + weight loss

Magnitude matches the **Retatrutide phase 2 trial** (NCT04881760; Jastreboff, NEJM 2023), which produced 17–24% weight loss with tightly-tracking lipid improvements. The –91 triglyceride drop is the signature of glucagon/GIP triple-agonism plus your energy-deficit state (leptin 0.4, insulin 2.5). ApoB → 69 confirms a genuine atherogenic-particle reduction, not a calc artifact. Cagrilintide + caloric restriction additive. Research compounds (SLU-PP-332, MOTS-c, 5-Amino-1MQ) have **no human lipid trials** — don't credit them.

- **HDL 76→52:** expected, benign — reciprocal of a collapsed-TG state during active rapid weight loss; usually settles at maintenance. With ApoB 69 / TG 37, risk profile is *better*, not worse.
- **Lp(a) 88→58: do not attribute to peptides** — Lp(a) is ~90% genetic and isn't lowered by incretins. Assay/lab noise across portals. Re-measure once on a single consistent assay.

D) Leukocyte esterase TRACE (urine) → clinically minor

Trace LE with **negative nitrite, protein, and blood, clear urine, asymptomatic** is most likely contamination or incidental low-grade pyuria. IDSA 2019 (Nicolle, PMID 30895288): don't screen/treat asymptomatic bacteriuria in men — **no antibiotics indicated**. If symptom-free, just repeat a clean-catch midstream sample at routine follow-up; pursue micro + culture only if persistent or symptomatic.

4. Suggested next steps (to bring to your physician)

Already pending — just get results: iron panel (TIBC/%sat) and testosterone/SHBG/free-T.

Cheap confirmatory adds that close every open question:

- **Reticulocyte count** — resolves the low-RBC question (diluted vs true anemia).
- **MMA + homocysteine, RBC folate, TSH** — confirms macrocytosis is benign/constitutional.
- **Single-assay Lp(a) recheck** — establish true genetic baseline; stop tracking cross-portal noise.
- **Repeat clean-catch UA** if you want to clear the trace LE (optional if asymptomatic).

Monitoring notes:

- Keep periodic CBCs — if GH agents are reduced, Hct could rise (TRT erythrocytosis surveillance).
- AST 44 is likely muscle — confirm by drawing after ≥48h rest from heavy training (consider CK/GGT).
- You're in an aggressive energy-deficit phase (leptin 0.4, falling HDL, very low TG) — protect lean mass with protein + resistance training through the retatrutide taper.

> **Honest summary.** Nothing in yesterday's results is alarming. Two flags are constitutional/longstanding, one is trivial, and the trend lines — lipids, glucose/insulin, inflammation, testosterone — are genuinely impressive and consistent with what your protocol is designed to do. The most “abnormal” CBC finding (low RBC despite TRT) is actually a signature of your GH/IGF + daily-SubQ-T design keeping you out of erythrocytosis.

Generated for Roman Semenov, June 9, 2026. Literature synthesis via PubMed and ClinicalTrials.gov. This document is informational and does not constitute medical advice or a diagnosis; review all findings and any testing decisions with your treating clinician.